Mehboob Alam

Las Cruces, NM | | mehboob@nmsu.edu
linkedin.com/in/mehboob-alam-530483185 | github.com/MehboobAlam92

PROFESSIONAL SUMMARY

M.S. Mathematics candidate (May 2027) with advanced Python programming skills and computational research experience in machine learning, numerical methods, optimization, and statistical modeling. Implemented Julia machine learning algorithms from scratch and evaluated regression generalization, clustering convergence, and image segmentation. Co-author of 18 published research articles in applied and computational mathematics.

TECHNICAL SKILLS

Programming: Python (advanced), Julia, MATLAB, Mathematica; Jupyter notebooks
Machine Learning: Regression (linear, multivariate, polynomial), L2 regularization, batch/online K-means, clustering, image segmentation, train/test model evaluation, root-mean-square error
Quantitative Methods: Numerical optimization, probability and statistics, linear algebra, numerical simulation, finite differences, radial basis function collocation, finite element analysis
Software: Abaqus, DEAFrontier, Microsoft Excel, LaTeX

SELECTED TECHNICAL PROJECTS

Regression, Model Selection, and Regularization *Julia | 2026*

- Implemented bivariate/multivariate linear regression and polynomial least-squares fitting; generated noisy linear and sinusoidal datasets for controlled experiments.
- Compared polynomial degrees 0–9 using training and test RMS error on a separate 500-sample simulated test dataset to investigate overfitting and generalization.
- Implemented L2-regularized polynomial regression and compared weak versus strong penalties to analyze model complexity and fit quality.

Unsupervised Learning, K-means, and Image Segmentation *Julia | 2026*

- Implemented batch and online K-means algorithms, nearest-centroid assignment, and synthetic multivariate Gaussian sampling in Julia.
- Evaluated centroid estimation across 100 random trials and convergence across 500 trials; observed 2–7 batch K-means iterations in the reported experiment.
- Applied clustering to RGB image color quantization and compared segmentation using 4, 8, and 16 representative colors.

Railway Resource Allocation Optimization *DEAFrontier | Operations Research*

- Modeled staffing allocation across 44 railway stations using Parabolic Data Envelopment Analysis and a sequential allocation algorithm.
- Improved modeled system efficiency from 56% to 99.5% through resource reallocation; investigated integer-valued staffing decisions.

Numerical Modeling of Time-Fractional PDEs *MATLAB | Scientific Computing*

- Applied meshfree radial basis function collocation and finite-difference time discretization to approximate nonlinear time-fractional partial differential equations.
- Evaluated numerical accuracy and theoretical temporal convergence order $O(\Delta t^{2-\alpha})$; published related research on the FitzHugh–Nagumo equation.

Python Machine Learning Notebooks *Python | Jupyter | GitHub*

- Published three Jupyter notebooks featuring introductory and intermediate scikit-learn exercises and missing-value imputation.

EDUCATION

New Mexico State University, Las Cruces, NM **Expected May 2027**
M.S. Mathematics | GPA: 3.5/4.0
Coursework: Machine Learning, Probability Theory and Applications, Statistics, Advanced Linear Algebra, Measure and Integration

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology, Pakistan **Jan 2024**
M.S. Engineering Sciences (Applied Mathematics) | GPA: 3.75/4.0

University of Peshawar, Pakistan **Nov 2020**
B.S. Mathematics | GPA: 3.72/4.0

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant Aug 2025 – Present
New Mexico State University | Las Cruces, NM

- Support differential equations and linear algebra coursework, assess quantitative solutions, and communicate technical feedback to students.
- Maintain grading records and organize weekly assessments using Canvas.

Lecturer, Mathematics and Statistics Feb 2024 – Mar 2025
University of Lahore | Sargodha, Pakistan

- Taught mathematical modeling, probability and statistics, differential equations, and other quantitative courses.
- Coordinated course timetables and supported departmental scheduling and academic operations.

Graduate Teaching Assistant Feb 2022 – Jan 2024
Ghulam Ishaq Khan Institute | Pakistan

- Supported engineering statistics, numerical analysis, differential equations, and calculus courses.

RESEARCH AND PROFESSIONAL SERVICE

- Co-authored 18 published research articles in applied mathematics, numerical methods, and differential equations, including research in *Fractal and Fractional*, *Alexandria Engineering Journal*, and *Chaos, Solitons & Fractals*.
- Served as a peer reviewer for journals including *Scientific Reports*, *Chaos, Solitons & Fractals*, and *Communications in Nonlinear Science and Numerical Simulation*.